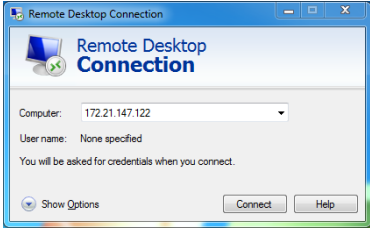
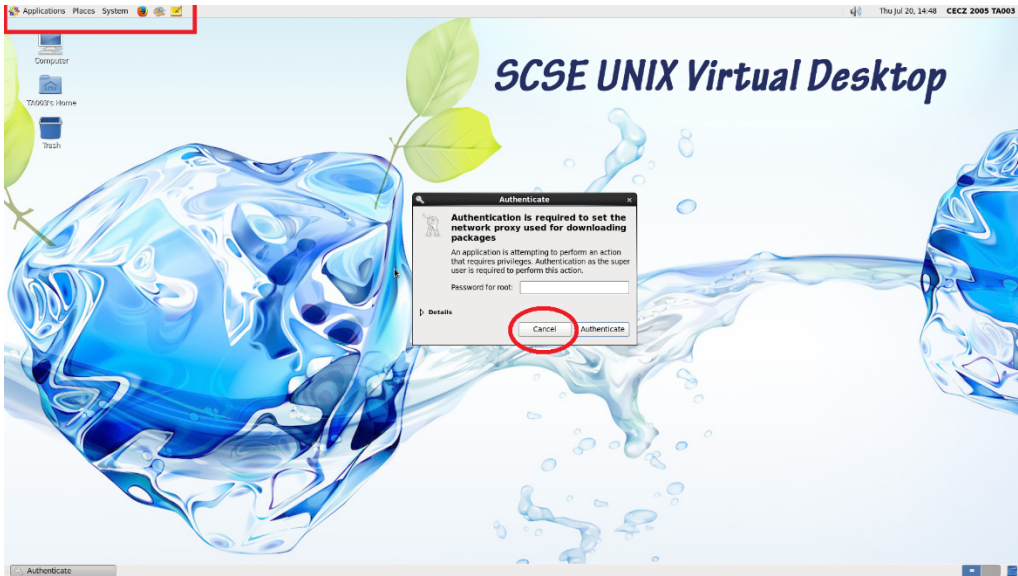


General Lab Guidelines for **SC2005** - Operating Systems

Lab sessions overview	1. Sign on the attendance sheet 2. Free Seating 3. Please backup your work regularly to external drive/ email etc.		
	Basically you need to remote access to one of the 40 Linux machines to compile/run your code		
	There are a total of 4 lab sessions : Lab 1 : Experiment 1 on Nachos Threads Lab 2 : Experiment 2 on CPU Scheduling Lab 3 : Experiment 3 on Process Synchronization Lab 4 : Experiment 4 on Virtual Memory		
	Software needed		
	1	Remote Desktop Connection	 To connect to your Linux workspace
	2	winscp	 For file transfer between PC and remote Linux machine
Linux Account	Username : NTU email account ID (If your email account is ntu123@e.ntu.edu.sg, your username will be NTU123 !!! (Username must be UPPERCASE)		
	Password : Welcome2SWL (Case Sensitive) Please CHANGE your password during your first Lab Issue yppasswd command within Linux environment To reset your password (If you forget your password) 1. Send an email to askgchia@ntu.edu.sg stating your group number; you must use your NTU email account to send so that we could verify your identity. 2. Or approach the technician with your matriculation card during Lab Your password will be reset to “Welcome2SWL”		
Free Access	Other than the free access slots allocated, please proceed to Software-Projects Lab (N4-B1b-11) for free access.		

Linux Env Login Info	1. Take note of the IP address assigned to your PC (The assigned IP address is labelled on the PC) Alternatively, refer to “SC2005 PC IP Assignment” section below and copy the corresponding IP address. It is advisable to use the assigned IP address to reduced traffic congestion. 2. Launch “Remote Desktop Connection” program and use the assigned IP address to connect  3. Please wait for the Linux desktop to fully load, you should see the menu bar. Please click “Cancel” on the [Authenticate] windows  4. Launch “Terminal” (For you to run command and compile codes) From Menu Bar, “Applications” -- “System Tools” -- “Terminal” 5. Follow your Lab manual to complete your lab 6. IMPORTANT !!, Please select “System” “Log Out xxxxx” from the menu bar when you have completed your lab work. Failure to do so might cause future login issue.
Remote Access from other Lab	1. Launch “Remote Desktop Connection” program and use any of the 40 IP addresses from 10.96.188.51 to 10.96.188.90. For load balancing purpose, please use a random IP, preferably to use your Lab attendance number as reference. 2. Continue your work (similar to lab)
Remote Access from Home	To remote access from home, you need to 1. use VPN to connect to NTU (only If you are outside NTU) VPNGate.ntu.edu.sg 2. Launch “Remote Desktop Connection” program and use any of the 40 IP addresses from 10.96.188.51 to 10.96.188.90. For load balancing purpose, please use a random IP, preferably to use your Lab attendance number as reference. 3. Continue your work (similar to lab)

Common command	make ----- to compile your code make clean ---- to delete all the already compiled object files ./nachos ----- to run your compiled nachos code ./nachos -d ---- to run your compiled nachos code with detail output ./nachos -d > file.txt --- to run your compiled code and output the result to a file name file.txt yppasswd ----- to change your account password
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SC2005 PC IP Assignment

For remote access, please choose a random IP address from the pool for better load balancing.

PC Number	IP address to use		PC Number	IP address to use
PC1	10.96.188.51		PC21	10.96.188.71
PC2	10.96.188.52		PC22	10.96.188.72
PC3	10.96.188.53		PC23	10.96.188.73
PC4	10.96.188.54		PC24	10.96.188.74
PC5	10.96.188.55		PC25	10.96.188.75
PC6	10.96.188.56		PC26	10.96.188.76
PC7	10.96.188.57		PC27	10.96.188.77
PC8	10.96.188.58		PC28	10.96.188.78
PC9	10.96.188.59		PC29	10.96.188.79
PC10	10.96.188.60		PC30	10.96.188.80
PC11	10.96.188.61		PC31	10.96.188.81
PC12	10.96.188.62		PC32	10.96.188.82
PC13	10.96.188.63		PC33	10.96.188.83
PC14	10.96.188.64		PC34	10.96.188.84
PC15	10.96.188.65		PC35	10.96.188.85
PC16	10.96.188.66		PC36	10.96.188.86
PC17	10.96.188.67		PC37	10.96.188.87
PC18	10.96.188.68		PC38	10.96.188.88
PC19	10.96.188.69		PC39	10.96.188.89
PC20	10.96.188.70		PC40	10.96.188.90

Optional

Optional

For information only

Alternative method to access your Linux environment

1. Refer to “**SC2005 PC IP Assignment**” section above and **copy the corresponding IP address**
It is advisable to use the IP address corresponding to the PC number which you are using to connect to the remote Linux machine.
2. Execute “**putty**” program
3. **Paste the IP address** into **putty**, leave the port as **22**, connection type is **SSH** and **click open**, select “**Yes**” when prompted
4. Login using
Username : NTU email account ID
(if your email address is ntu123@e.ntu.edu.sg, your username will be **NTU123**)

You do not need to use this if you are using "Remote Desktop Connection" to connect to the Linux workspace

!!! (Username must be **UPPERCASE)**

Password : **Welcome2SWL (Case sensitive)**

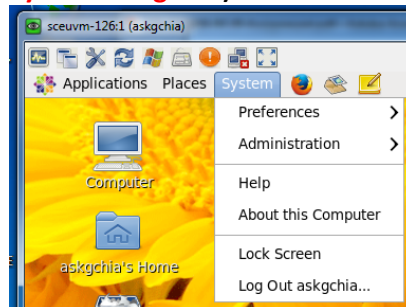
If you are comfortable with command line mode, you can do all your labs using command mode, it will be much faster and you do not need VNC (You do not need step 5 onwards)

Step 5 onwards allows you to connect to the remote Linux machine in graphical mode.

5. Type the following command in **putty** to start vncserver for GUI connection
vncserver -geometry 1600x900
6. **!!Take note** on the display number that appear after issuing vncserver command (eg sceuvm-121:**1**, you will need it later in step 10, the **:1** is the display number)
7. **Choose a VNC password**, you need to retype password again to confirm, if you need to change your VNC password later, you could issue "**vncpasswd**" command
8. **Run "UltraVNC viewer"** (on desktop)
Use the IP address from **step 1** and display number from **step 8** to connect
Eg. 10.96.188.51:1
9. Use the password in **step 9** when prompted
10. Continue your lab, you can only compile/run your code in "Terminal" (To open Terminal : select Applications, System Tools and finally Terminal)

11. IMPORTANT !!

You need to **logout** your remote GUI session when you finish your lab (**Select System – Log out**)



common VNC command

vncpasswd ----- to change your VNC password

vncserver -geometry 1600x900 --- to start a VNC session with display size of 1600x900

vncserver -list ---- to list currently running VNC session